**8.**

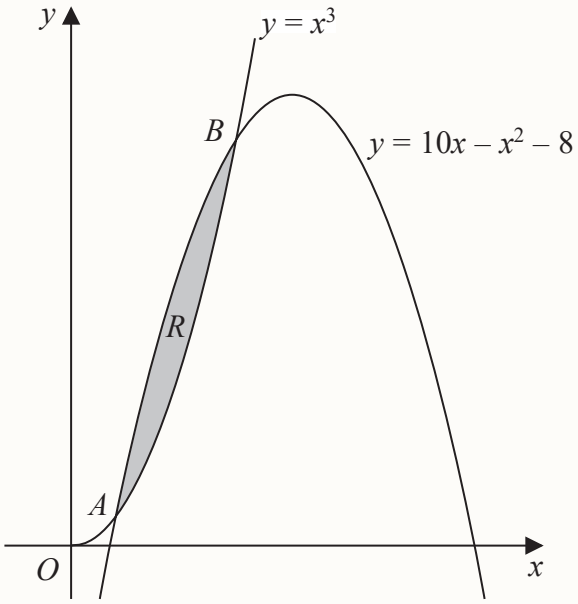



**Figure 2**

Figure 2 shows a sketch of part of the curves $C_1$ and $C_2$ with equations

$$C_1: y = 10x - x^2 - 8 \qquad x > 0$$

$$C_2: y = x^3 \qquad x > 0$$

The curves $C_1$ and $C_2$ intersect at the points $A$ and $B$.

(a) Verify that the point $A$ has coordinates (1, 1)

**(1)**

(b) Use algebra to find the coordinates of the point $B$

**(6)**

The finite region $R$ is bounded by $C_1$ and $C_2$

(c) Use calculus to find the exact area of $R$

**(5)**